

6 (a) $PV = nRT$

$$\begin{aligned} n &= \frac{PV}{RT} \\ &= \frac{(4.5 \times 10^5)(0.075)}{(8.31)(273 + 60)} \quad [\text{C1}] \\ &= 12.2 \text{ mol} \quad [\text{A1}] \end{aligned}$$

(b) (i) since pressure, n and R are constant,

$$\begin{aligned} \frac{V_1}{T_1} &= \frac{V_2}{T_2} \\ V_2 &= \frac{V_1}{T_1} T_2 = \frac{0.075}{333} \times (150 + 273) \quad [\text{C1}] \\ &= 0.0953 \text{ m}^3 \quad [\text{A1}] \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \Delta U &= \frac{3}{2} nR\Delta T \\ &= \frac{3}{2} (12.2)(8.31)(150 - 60) \quad [\text{C1}] \\ &= 1.37 \times 10^4 \text{ J} \quad [\text{A1}] \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad WD_{by} &= P\Delta V = (4.5 \times 10^5)(0.0953 - 0.075) \quad [\text{C1}] \\ &= 9.14 \times 10^3 \text{ J} \quad [\text{A1}] \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad \Delta U &= q + w \\ q &= \Delta U - w \\ &= 1.37 \times 10^4 - (-9.14 \times 10^3) \quad [\text{C1}] \\ &= 2.28 \times 10^4 \text{ J} \quad [\text{A1}] \end{aligned}$$

